Abhinav Krishna Kandukuri

PES1UG24CS701

1	1 A	./test_objects output showing all tests passing abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ ./test_objects Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5 Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed02 afc12 PASS: blob storage PASS: deduplication PASS: integrity check All Phase 1 tests passed. abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
1	1 B	find .pes/objects -type f showing sharded directory structure abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ ./test_objects Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5 Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed02 afc12 PASS: blob storage PASS: deduplication PASS: integrity check All Phase 1 tests passed. abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ find .pes/objects -type f .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaf .pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566 abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
2	2 A	./test_tree output showing all tests passing abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ ./test_tree Serialized tree: 139 bytes PASS: tree serialize/parse roundtrip PASS: tree deterministic serialization All Phase 2 tests passed. abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
2	2 B	xxd of a raw tree object (first 20 lines)

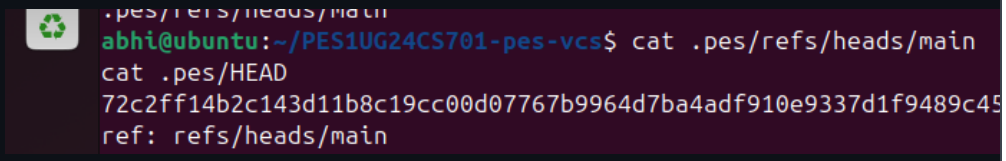
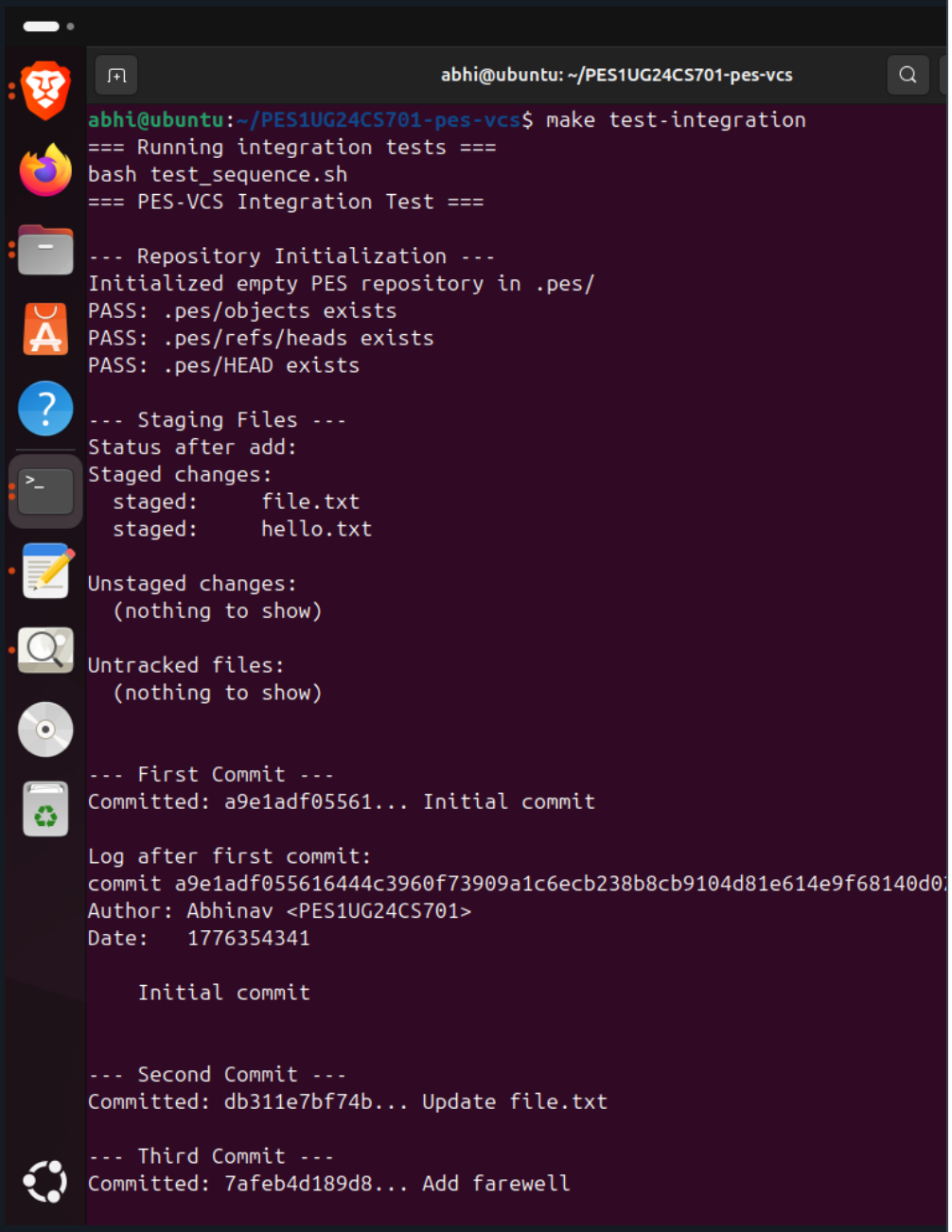
		0 such file or directory abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ find .pes/objects -type f | head .pes/objects/e9/bc3cb828854372249c5d6ed2ddc22cbb0fd466c1e33cd32c103d93 abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ xxd .pes/objects/e9/bc3cb82885437 22cbb0fd466c1e33cd32c103d93b925c9a2 | head -20 00000000: 636f 6d6d 6974 2031 3730 0074 7265 6520 commit 170.tree 00000010: 3665 6631 3962 3431 3232 3563 3533 3639 6ef19b41225c5369 00000020: 6631 6331 3034 6434 3564 3864 3835 6566 f1c104d45d8d85ef 00000030: 6139 6230 3537 6235 3362 3134 6234 6239 a9b057b53b14b4b9 00000040: 6239 3339 6464 3734 6465 6363 3533 3231 b939dd74decc5321 00000050: 0a61 7574 686f 7220 4162 6869 6e61 7620 .author Abhinav 00000060: 3c50 4553 3155 4732 3443 5337 3031 3e20 <PES1UG24CS701> 00000070: 3137 3736 3335 3335 3035 0a63 6f6d 6d69 1776353505.commi 00000080: 7474 6572 2041 6268 696e 6176 203c 5045 tter Abhinav <PE 00000090: 5331 5547 3234 4353 3730 313e 2031 3737 S1UG24CS701> 177 000000a0: 3633 5353 3530 350a 0a66 6972 7374 2063 6353505..first c 000000b0: 6f6d 6d69 74 ommit abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
3	3 A	pes init → pes add → pes status sequence abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ ./pes init echo "hello" > a.txt ./pes add a.txt ./pes status Initialized empty PES repository in .pes/ Staged changes: staged: a.txt Unstaged changes: (nothing to show) Untracked files: untracked: tree.h untracked: commit.c untracked: test_sequence.sh untracked: index.h untracked: object.c untracked: README.md untracked: commit.h untracked: index.c untracked: test_objects.c untracked: tree.c untracked: Makefile untracked: test_tree.c untracked: pes.h untracked: .gitignore untracked: pes.c abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
3	3 B	cat .pes/index showing the text-format index untracked: pes.h untracked: .gitignore untracked: pes.c abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ cat .pes/index 100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 17 abhi@ubuntu:~/PES1UG24CS701-pes-vcs$

4	4 A	pes log output with three commits

```
00000090: 5331 5547 3234 4353 3730 313e 2031 3737  S1UG24CS701> 177
000000a0: 3633 3533 3530 350a 0a66 6972 7374 2063  6353505..first c
000000b0: 6f6d 6d69 74                             ommit
abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ rm -rf .pes
./pes init
export PES_AUTHOR="Abhinav <PES1UG24CS701>"

echo "hello" > a.txt
./pes add a.txt

./pes commit -m "first"
./pes commit -m "second"
./pes commit -m "third"

./pes log
Initialized empty PES repository in .pes/
Committed: 43e66c42aba8... first
Committed: 9a9cf3a23d70... second
Committed: 72c2ff14b2c1... third
commit 72c2ff14b2c143d11b8c19cc00d07767b9964d7ba4adf910e9337d1f9489c4
Author: Abhinav <PES1UG24CS701>
Date:   1776354268

    third

commit 9a9cf3a23d7013adfe7338c1f8782ea7b0ce9078e433644265784eb520bff5
Author: Abhinav <PES1UG24CS701>
Date:   1776354268

    second268

third

commit 43e66c42aba81552600820f31124c4b3a184a71de862c09de51e13ed3a194a
Author: Abhinav <PES1UG24CS701>
Date:   1776354268

    first4268
a

abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
```

4	4 B	find .pes -type f | sort showing objec growth

```
abhi@ubuntu:~/PES1UG24CS701-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d3376390
.pes/objects/43/e66c42aba81552600820f31124c4b3a184a71de862c09de5
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b9
.pes/objects/72/c2ff14b2c143d11b8c19cc00d07767b9964d7ba4adf910e9
.pes/objects/9a/9cf3a23d7013adfe7338c1f8782ea7b0ce9078e433644265
.pes/refs/heads/main
```

4	4 C	cat .pes/refs/heads/main and cat .pes/HEAD 
Fin al	--	Full integration test (make test-integration) 

abhi@ubuntu: ~/PES1UG24CS701-pes-vcs

```
--- Second Commit ---
Committed: db311e7bf74b... Update file.txt

--- Third Commit ---
Committed: 7afeb4d189d8... Add farewell

--- Full History ---
commit 7afeb4d189d8e9e8aaaf51563fac115cee4e1e2363bb6e61f35064016d48b406
Author: Abhinav <PES1UG24CS701>
Date:   1776354341

    Add farewell

commit db311e7bf74bbbc46ba65d457af5253ca128f7a1c75a71460ecc8b89c97dc2b2
Author: Abhinav <PES1UG24CS701>
Date:   1776354341

    Update file.txt

commit a9e1adf055616444c3960f73909a1c6ecb238b8cb9104d81e614e9f68140d026
Author: Abhinav <PES1UG24CS701>
Date:   1776354341

    Initial committial commitEQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
7afeb4d189d8e9e8aaaf51563fac115cee4e1e2363bb6e61f35064016d48b406

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/7a/feb4d189d8e9e8aaaf51563fac115cee4e1e2363bb6e61f35064016d48b406
.pes/objects/a9/e1adf055616444c3960f73909a1c6ecb238b8cb9104d81e614e9f68140d026
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
```

```
--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/7a/feb4d189d8e9e8aaaf51563fac115cee4e1e2363bb6e61f35064016d48b406
.pes/objects/a9/e1adf055616444c3960f73909a1c6ecb238b8cb9104d81e614e9f68140d026
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/db/311e7bf74bbbc46ba65d457af5253ca128f7a1c75a71460ecc8b89c97dc2b2

=== All integration tests completed ===
abhi@ubuntu:~/PES1UG24CS701-pes-vcs$
```

Analysis Questions:

Q5.1 — pes checkout

- Update .pes/HEAD to point to target branch
- Read commit hash from .pes/refs/heads/<branch>
- Load commit → get tree
- Update working directory to match tree (add/remove/overwrite files)
- Update .pes/index accordingly

Complexity:

- Must avoid overwriting uncommitted changes
- Need to keep **HEAD, index, working directory** consistent

Q5.2 — Dirty working directory detection

For each tracked file:

- Compare working file hash with index hash → detect modification
- Compare index hash with target branch tree hash → detect difference

Conflict if:

- File modified locally AND differs in target branch

→ Refuse checkout

Q5.3 — Detached HEAD

- HEAD stores commit hash directly (no branch)
- New commits are created but **not linked to any branch**
- These commits can be garbage collected

Recovery:

- Create a branch pointing to current commit

Q6.1 — Garbage collection

Algorithm:

1. Start from all branch heads
2. Traverse commits → trees → blobs
3. Store visited hashes in a **hash set**
4. Delete objects not in set

Scale:

- ~100k commits → ~500k–700k objects visited

Q6.2 — GC race condition

- Commit creates objects (not yet referenced)
- GC runs and deletes them (seen as unreachable)
- Commit updates branch → now references missing objects

Fix (Git):

- Atomic updates
- Locking
- Delay deletion (grace period / reflog)